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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 01/08/2007 |
| **PI Name:** | Dinges, David F. Ph.D. | | |
| **Project Title:** | Optical Computer Recognition of Performance Under Stress | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Neurobehavioral and Psychosocial Factors Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **BHP**:Behavioral Health & Performance (archival in 2017) | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 19104-4209 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Biomedical Research & Countermeasures 03-OBPR-04 |
| **Start Date:** | 03/01/2004 | **End Date:** | 02/28/2008 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 57 |
| **No. of Bachelor's Candidates:** | 51 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Metaxas, Dimitris ( Rutgers University ) | | |
| **Grant/Contract No.:** | NCC 9-58-NBPF00402 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | (1) This project is focused on the early objective detection of neurobehavioral problems in space flight. The overarching goal is to develop an unobtrusive, automated optical technology to detect psychological distress when astronauts are working during spaceflight. The primary aim is to determine whether optical computer recognition algorithms based on changes in facial expressions can discriminate behavioral stress induced by low versus high workload performance demands.<br>Astronauts are required to perform mission-critical tasks at a high level of functional capability throughout spaceflight. There are a number of stressors that can compromise their ability to do so. In order to provide timely countermeasures for stressor-induced impairments in astronauts, objective, unobtrusive measures of the presence of stress reactions—especially during performance demands—are needed. This project seeks to achieve such a measurement | | |

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| **Task Description:** | methodology, through continued collaboration between two laboratories: one with expertise in the evaluation of behavioral and physiological responses to stressful performance conditions (D. Dinges, Unit for Experimental Psychiatry, University of Pennsylvania School of Medicine), and the other with expertise in optical computer recognition of human expressions and gestures (D. Metaxas, Computational Biomedicine Imaging and Modeling Center, Rutgers University). The goal of this collaboration is to further develop and test an optically-based computer recognition algorithm of the face to reliably detect the presence of stress. The proposed computer-based recognition system builds on the research of Prof. Metaxas by utilizing automatic optical tracking of human subjects' subtle anatomical and motoric changes in facial expressions. Video input to the system is provided from experiments performed in the laboratory of Prof. Dinges. In this project we are experimentally testing the extent to which optical computer recognition algorithms based on facial expressions can detect behavioral stress during cognitive performance, both with and without facial edema. Because the facial edema that occurs in microgravity due to fluid shifts may degrade optical computer recognition of low versus high stress induced by performance demands, we are testing the following hypothesis twice in the same subjects, under both normal (no facial edema) conditions and under facial edema conditions. Hypothesis: Optical computer recognition algorithms of the face can reliably discriminate when subjects are undergoing behavioral stress associated with low versus high cognitive workload demands. In addition to testing this hypothesis, we are evaluating the extent to which the hypothesized outcome is influenced by the following potentially critical modifiers: Age; gender; ethnic background; degree of alexithymia (i.e., difficulty in recognizing and describing one’s emotions); degree of subjective distress and physiological response (i.e., heart rate) during high workload performance demands; and the level of performance during difficult cognitive tasks. It is noteworthy that the test of this hypothesis allows optical algorithms to be idiosyncratic to individual subjects (i.e., no requirement of algorithm generalizability across subjects).<br><br>We are also determining if optical computer recognition algorithms of facial expressions can be automated for online, real-time use in facial detection of behavioral stress during performance demands.<br><br>We are also establishing the effects of low versus high behavioral stress induced by differential workload demands on performance on the WinSCAT, a cognitive test battery used by NASA JSC during space missions. This specific aim is exploratory and hypothesis-generating.<br><br>A single-blind, repeated-measures cross-over trial is being used to achieve the specific aims and provide data required to test the hypotheses. A total of n=80 adults will be studied in a day-long laboratory session. The procedures for this experiment were completed in the past and preparations are underway for data acquisition in the coming year.<br><br>(2) During the last year, substantial progress has been made on increasing the sophistication of the optical computer recognition system to detect facial changes during performance stress. Significant advances in programming and enhancement of the capabilities of the computer recognition algorithm were completed, to develop a technique for robust 3-dimensional (3D) tracking of facial expression. Using this approach, numerous low-level, well-known computer vision algorithms were initially used to extract 2-dimensional (2D) information (tracking single points, edges, shading and optical flow), from video-recording of each subject's face during low and high workload performance demands. Put simply, the new technique, developed by Dr. Metaxas and colleagues, allowed the translation of 2D video footage to a form that could be used for tracking the 3D orientation and translation of the face, as well as parameters that describe the movement of eyebrows, mouth, etc. This novel technology provided us with a sensitive and unobtrusive mechanism for the measurement of changes in facial expression. Data analyses of the accuracy of the algorithm to detect performance stress are continuing on the first n=60 subjects studied. We are continuing with the development of the deformable face masks to increase the quality and precision of the tracking.<br><br>(3) The findings to date continue to support the hypothesis that an Optical Computer Recognition system can be developed to detect behavioral stress from facial expressions.<br><br>(4) In the coming year, data acquisition will begin on experiment 2. Work will also continue on improving the Optical Computer Recognition algorithm by adding facial touching gestures. Finally, work will continue on automated application of the deformable mask from video of the moving face to increase practical applicability. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The study focuses on the ability of an unobtrusive, automated optical technology to detect psychological distress (and the need for countermeasures for it) during operational performance. The knowledge gained has the potential to identify an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction due to work-related stress in spaceflight and in many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel. |
| **Task Progress:** | The overarching goal of the project is to provide an unobtrusive objective detection of stress during performance in space flight using machine vision (computational model-based tracker). This technology is being developed in experiments in which optical tracking of healthy adults’ subtle anatomical and motor changes in facial expressions occurs during exposure to varying degrees of workload-based behavioral stressors. A second experiment—approved by the University of Pennsylvania Institutional Review Board—involves expanding the stressor conditions to include the sleep deprivation and affective probes. Affective states are being experimentally manipulated using high intensity digital photographs from the International Affective Picture System of both positive and negative valence. Subjects’ responses to the probes are measured by self report (POMS), heart rate, and startle response. The startle probe being used is validated in extensive research on affect regulation. It consists of a 90dB burst of white noise that elicits an involuntary eye-blink in the subject. The magnitude of the response gives an objective indication of the affective state the subject experiences. Heart rate is also measured continuously throughout the affective probes, which gives an objective physiological correlate of affective intensity. The behavioral stress paradigm, which uses performance difficulty and negative informational and social feedback, has been maintained in the form validated with the first 60 subjects.<br>The computerized deformable mask of the face has been successfully automated by the Metaxas laboratory to track head and face movements in real time. The cue integration-based tracking system accurately captures rigid and non-rigid parameters of different parts of the face (eyebrows, lips). The labeled sequences are then used to train the recognition system, which consists of generative (Hidden Markov Model) and discriminative (Support Vector Machine) parts that yield results superior to using either approach individually. Accuracy and practical feasibility of the technique is being |

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| | improved further with automatic multi-resolution selection for the discretization of the mask, and automated face detection and mask initialization algorithms. Data have been collected in the current experiment on 18 healthy adults undergoing the behavioral stressors. Video footage has been sent to the Metaxas lab to be analyzed using the most recent version of the OCR algorithm. |
| **Bibliography Type:** | Description: (Last Updated: 06/30/2026) |
| **Articles in Peer-reviewed Journals** | Dinges DF, Rider RL, Dorrian J, McGlinchey EL, Rogers NL, Cizman Z, Goldenstein SK, Vogler C, Venkataraman S, Metaxas DN. "Optical computer recognition of facial expressions associated with stress induced by performance demands." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B172-82. PMID: 15943210 , Jun-2005 |
| **Articles in Peer-reviewed Journals** | Dinges DF, Venkataraman S, Mc Glinchey EL, Metaxas DN. "Monitoring of facial stress during space flight: Optical computer recognition combining discriminative and generative methods." Acta Astronautica, in press 2006. , Jan-2006 |
| **Awards** | "David F. Dinges: William C. Dement Academic Achievement Award, American Academy of Sleep Medicine." Jan-2006 |